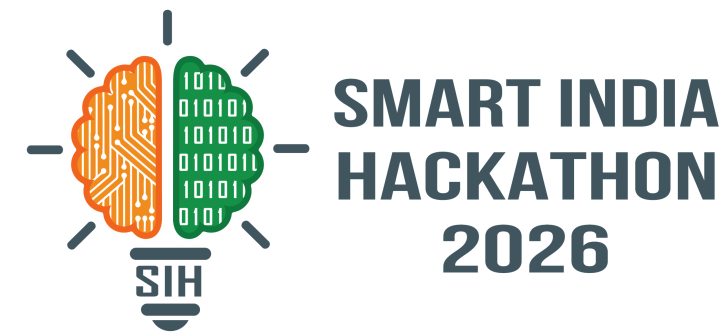
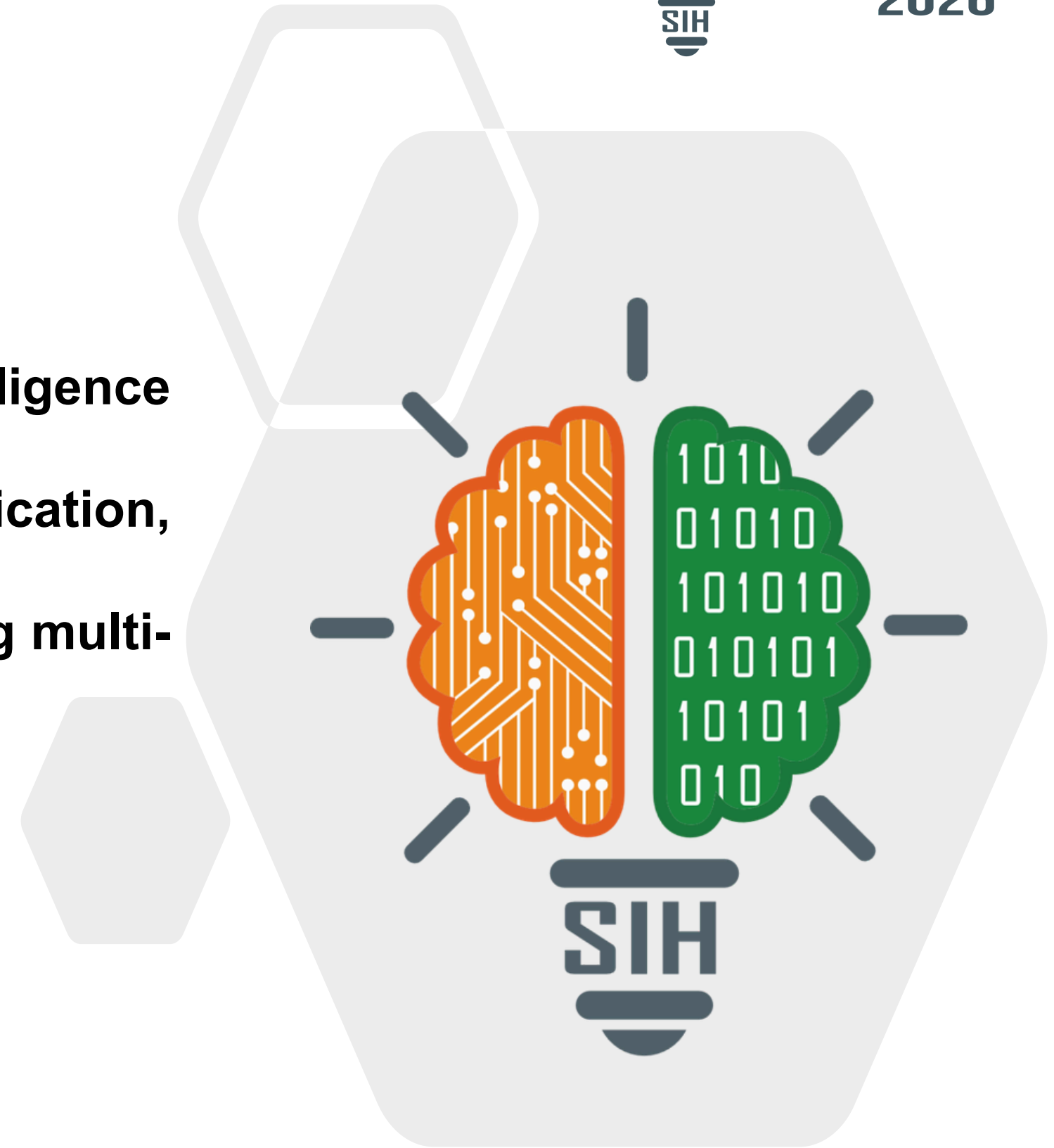




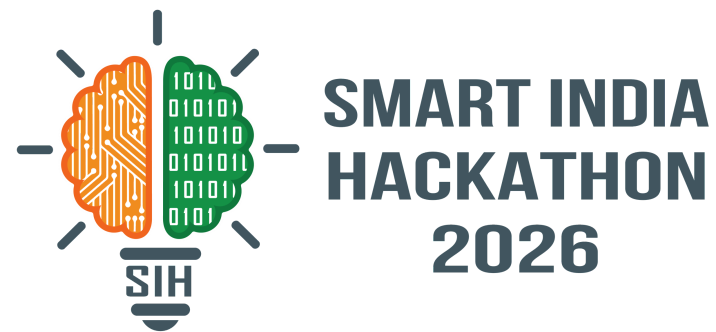
SMART INDIA HACKATHON 2026



- **Problem Statement ID – SIH26070**
- **Problem Statement Title:** To develop an Artificial Intelligence (AI)/Machine Learning (ML)-based system for the identification, classification, and prediction of tropical cyclone patterns using multi-source satellite data.
- **Theme: Disaster Management**
- **Team ID- 120**
- **Team Name – StrawHats**



CycloneWatch: AI-Powered Tropical Cyclone Tracker



Feature Aspect	Traditional NWP Models (IMD)	 CycloneWatch (Our Solution)
 Speed & Reaction	Highly accurate but slow to compute (takes hours).	Lightning fast: 12ms inference per satellite frame.
 Rapid Intensification	Struggles to predict RI reliably, causing warning delays.	Detects RI structural signatures instantly (Early Warning).
 Anomaly Detection	Breaks down at low latitudes (e.g., Ockhi 2017).	Reads visual patterns, ignoring latitude constraints.
 Interpretation	Human meteorologists manually review structural evolution.	Automates Dvorak Technique structural classification.
 Data Processing	Complex equations requiring Supercomputers.	Compact 540k parameter CNN running on standard CPUs.

TECHNICAL APPROACH

Data Ingestion

Collecting satellite data from NOAA and ISRO to ensure continuous updates.

API Backend

Using FastAPI and SQLite storage for efficient data management and retrieval.

ML Processing

Utilizing CNN and ConvLSTM networks to classify data and recognize patterns.

UI Dashboard

React map providing an interactive interface for decision support.

Technology Stack

[FRONTEND UI]

  React.js, Leaflet (Map), Zustand

[BACKEND API]

  FastAPI, SQLite (Precomputations)

[AI/ML BRAIN]

 PyTorch PyTorch, Deep CNN, ConvLSTM

[DATA PIPELINE]

  Python, NetCDF4, AWS S3

Model Performance

[STRUCTURAL CLASSIFICATION]

78.3% accuracy across 5 distinct morphological classes.

[CENTER TRACKING (T+12h)]

Mean Absolute Error (MAE) of ~234 km. Beats our 255 km persistence baseline.

[INFERENCE SPEED]

Real-time: ~12 milliseconds per satellite frame on standard CPU.

FEASIBILITY, VIABILITY & FUTURE TECH

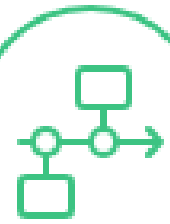


SMART INDIA
HACKATHON
2026



Assess Model Feasibility

- Utilizes free, accessible public satellite data from NOAA GridSat-B1.
- Built on proven open-source deep learning frameworks like PyTorch.
- Fully deployable without the need for massive cloud infrastructure.
- Prototype pipeline successfully tested on seven historical cyclones.



Plan Future Implementation

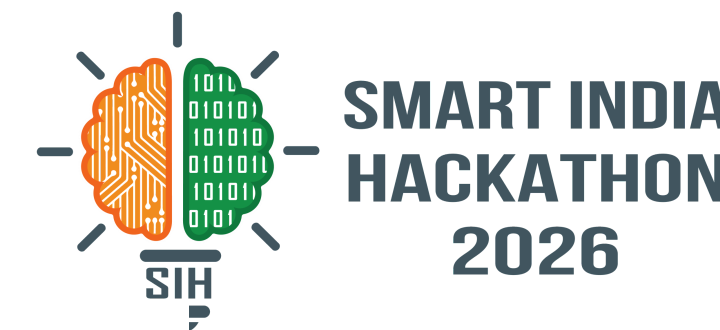
- Phase 2: Scale data resolution to 1km using ISRO MOSDAC / INSAT-3DR feeds.
- Phase 3: Transition to advanced ConvLSTM / GRU networks for non-linear predictions.
- Phase 4: Operational de-ployment to alert duty meteorologists.
- Reduce position Mean Absolute Error from 250km to less than 100km.



Evaluate Model Viability

- Lightning-fast inference times of just 12 milliseconds.
- Extensible modular API built with FastAPI for easy integration.
- Precomputed historical databases provide zero-latency access.
- Lightweight frontend developed using React is scalable.

IMPACT AND BENEFITS



Eliminating the 'Interpretation Gap'

Algorithms don't experience fatigue. CycloneWatch scans every new frame the instant it arrives, finding structural threat signatures hours before manual reviews.

Lead Time on Rapid Intensification

For Cyclone Ockhi, we gained a +36 hour lead time. For Tauktae, +30 hours. This is the difference between safe evacuation and mass casualty.

Significant Economic Protection

A single cyclone causes ₹10,000–50,000 crores in damages. An extra 12-hour warning window enables massive asset securing and better resource deployment.

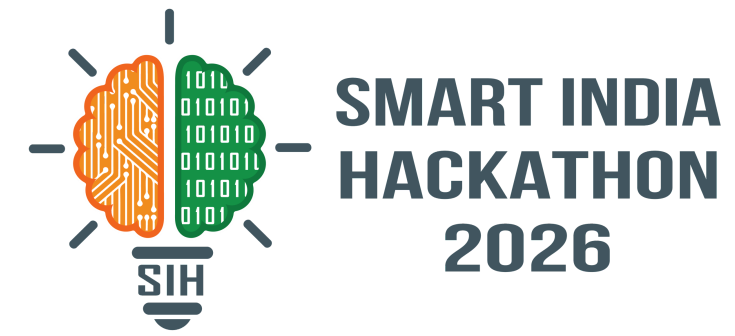
Unbiased Intensity Assessment

The AI validates structural signatures based strictly on visual data training, removing human subjectivity from intensity estimation.

An Early-Warning Safety Net

It does not replace IMD; it provides a '36-hour head start alert' to meteorologists before supercomputer equations complete their run.

RESEARCH AND REFERENCES



Primary Data Sources & Archives

- NOAA GridSat-B1: High-frequency, long-term global satellite data (4km resolution).
- IBTrACS (NOAA/WMO): The global best-track standard used as our ground-truth labels.
- NASA GIBS: Providing real-time historical visual cloud layers for our interactive map.

Live Meteorological Feeds

- Open-Meteo API: For real-time ocean current, wave height, SST, and wind speed data.

Meteorological Research & Taxonomies

- Dvorak Technique: Standardized visual structural pattern definitions.
- IMD Post-Cyclone Reports: Gap case analysis (e.g., Ockhi 2017, Tauktae 2021) to understand NWP delays.

Upcoming Integrations

- MOSDAC / INSAT-3DR (ISRO): Transitioning to 1km high-res datasets for Phase 2 implementation.